LESSON 1

STOCHASTIC VOLATILITY: HESTON

Derivative Pricing - Module 7

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Outline

- Intro to Stochastic Volatility
- The Heston model
- Monte-Carlo methods for Heston

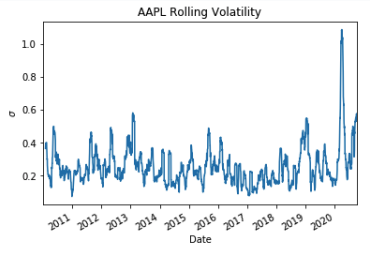
Stochastic Volatility

→ In past modules, we extended Black-Scholes considering models that allowed for some elasticity of variance (**local volatility**).

→ While this helped towards replicating the **volatility smile**, it is not enough.

→ In this module, we will introduce **stochastic volatility** models, specifically Heston (1993).

- Heston incorporates **2 important features**: Volatility is **(i) stochastic** and **(ii) mean-reverting**.



Heston Model

The Heston model comprises, essentially, two SDEs: one for the asset (Eq. 1) and one for its variance (Eq. 2):

$$dS = \mu S dt + \sqrt{\nu(t)} S dZ_1 \tag{1}$$

$$d\nu(t) = \kappa\left(\theta - \nu(t)\right) dt + \sigma\sqrt{\nu(t)} dZ_2 \tag{2}$$

where dZ_1 and dZ_2 are Wiener processes with correlation ρ.

- In Equation 1 (**SDE for asset**), under a risk-neutral setting, we'll have $\mu = r$.
- Equation 2 (**SDE for variance**) is what is new to us (is it though? $\rightarrow$ remember Vasicek?)
  - θ, long run average variance.
  - κ, rate of mean-reversion.
  - σ, volatility of volatility (vol of vol).

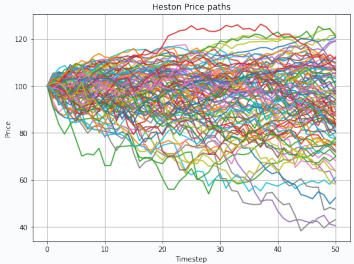
Simulating Heston

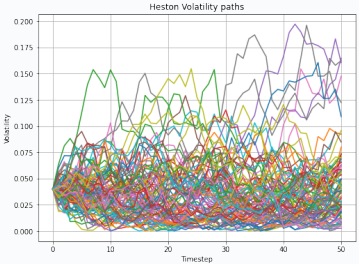
As before, given some parameters, we can **simulate Heston under Monte-Carlo** methods:

$$S_t = S_{t-1} e^{\left(r - \frac{\nu_t}{2}\right) dt + \sigma \sqrt{\nu_t} dZ_1}$$

$$\nu_t = \nu_{t-1} + \kappa \left(\theta - \nu_{t-1}\right) dt + \sigma \sqrt{\nu_{t-1}} dZ_2$$

One important input here will be the different parameters. For now, let's assume some and use MC to simulate:





Summary of Lesson 1

In Lesson 1, we have looked at:

- Stochastic volatility
- The Heston (1993) stochastic volatility model

⇒ **TO DO NEXT**: Now, please go to the associated Jupyter notebook for this lesson for a full development in Python of the Heston model under Monte-Carlo methods.

⇒ In the next lesson, we will take a closer look at option pricing in the Heston framework.

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